SPhO 2024 Experiment

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1 Problem Statement

This is recreated from memory - some small details may be inaccurate.

Materials Given:

1. 1 solid acrylic cylinder of radius R
2. 1 hollow acrylic cylinder of inner radius r, outer radius R
3. 1 solid acrylic cylinder of radius r
4. 1 solid brass cylinder of radius r
5. 1 solid aluminium cylinder of radius r
6. 1 solid PVC cylinder of radius r
7. 1 brass cube of known mass $m = (8.4 \pm 0.1)\,\mathrm{g}$
8. 1 wooden ramp at a fixed angle to the horizontal
9. 1 Jenga block
10. 1 stopwatch
11. 1 tape measure
12. 1 Vernier caliper

You are not allowed to use a weighing scale, drop any object into free-fall, or use torque balance in this experiment. Your methods should involve rolling.

Part A: Design an experiment to determine the mass of items 1-6, **using the stopwatch for most, if not all your steps.**

Part B: Design an experiment to determine the mass of items 1-6, **without using the stopwatch at all.**

Part C: State and explain some sources of experimental error in Parts A and B. Which of your two methods is more accurate in determining the masses? Explain why.